

Zhongyin Wang

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Education

University of Illinois Urbana-Champaign

Urbana, IL, USA

Master of Science in Information Management (GPA: 4.0/4.0)

Aug. 2025 – May 2027

- **Coursework:** Database Systems, Information Modeling, Software Engineering I, Statistical Models and Information.

Zhejiang University

Hangzhou, China

Bachelor in Civil Engineering (Smart Transportation & Systems) (GPA: 3.6/4.0)

Sept. 2020 – July 2024

- **Awards:** Zhejiang University Excellent Graduate, Structural Design Competition Second Prize.
- **Coursework:** Data Structures and Algorithm Analysis, C Programming.

Research Experience & Selected Projects

IlliniSphere: AI-assisted Research Pathway Discovery

May 2026 – Present

Research Assistant / Advisor: Prof. Yun Huang (University of Illinois Urbana-Champaign)

- Contributing to the design and prototyping of an AI-assisted platform for research pathway discovery and collaborative team formation.
- Developing LLM-based workflows that organize student interests, skills, and coursework into structured research pathway and team recommendations.
- Exploring AI-agent collaboration strategies to support team formation and decision-making through iterative system design and evaluation.

Nexus: LLM-based Academic Planning System

Mar. 2026 – Present

- Developed an intelligent scheduling system that automates the parsing of heterogeneous syllabus data (PDF/JSON) from Canvas and Coursera using **LLMs** for semantic extraction.
- Engineered a conflict-detection algorithm to generate optimized study plans, utilizing Redis caching for course metadata to ensure low-latency API responses.
- Designed an LLM-based agent workflow using Spring AI to coordinate task decomposition and personalized schedule generation.

Enterprise Compliance & Rewards Platform

Nov. 2025 – Jan. 2026

- Developed a graph-based data model using **Neo4j** and optimized Cypher queries for efficient relationship querying across hierarchical compliance policies and heterogeneous entities.
- Implemented a **Redis Cache-Aside** pattern to reduce MySQL database load by 30% while ensuring ACID transaction integrity.
- Designed RESTful APIs with JUnit/Mockito testing and containerized the system using Docker.

LLM-based Multi-table Learning for Credit Risk Modeling

Sept. 2024 – Dec. 2024

Research Assistant / Advisor: Prof. Yang Yang (Zhejiang University)

- Replicated the downstream regression task from the paper *Towards Cross-Table Masked Pretraining for Web Data Mining* to validate the effectiveness of cross-table pretraining.
- Extended the original row-level formulation to support table-level regression and classification on heterogeneous multi-table datasets.
- Implemented data preprocessing and sampling pipelines for cross-table representation learning across structured financial datasets.
- Evaluated model performance under different sampling strategies to support credit risk prediction.

Transformer-based Urban Road Network Design (Generative AI)

Dec. 2023 – June 2024

Bachelor's Thesis / Advisor: Prof. Hongsheng Qi (Zhejiang University)

- Developed a generative AI pipeline using **Transformer** and **Graph Neural Networks (GNN)** to automate the design of urban road networks from zero-to-one.
- Processed large-scale geospatial data via OSM APIs, converting raw geographic nodes into optimized **graph-based topologies** to evaluate connectivity and network resilience.
- Validated model performance by comparing generated layouts with existing traffic networks to support autonomous driving simulations.

Smart Tunnel Dynamic Emergency Response System

May 2022 – May 2023

Project Leader / Advisor: Prof. Ke Wu (Zhejiang University)

- Integrated tunnel fire dynamics models with multi-sensor data fusion to predict fire temperature fields and smoke spread in real-time.
- Leveraged **YOLO** algorithms and computer vision to automate fire source detection and scale estimation, reconstructing fire scenes based on physical dynamics equations.
- Developed a dynamic response platform using **Java** that allows real-time emergency coordination and response for critical infrastructure safety.

Industry Experience

IDX Exchange | Data Analyst InternBoise, ID (Remote) | Mar. 2026 – June 2026

- Built automated Python ETL pipelines to aggregate MLS transaction datasets; integrated FRED API for time-series analysis and engineered market intelligence metrics to support real estate decision-making.

Baidu, Inc. | NLP Algorithm Engineer InternBeijing, China | Apr. 2024 – June 2024

- Evaluated representative large language models through API-based experiments, analyzing inference latency, token efficiency, and long-context performance across multiple evaluation metrics.
- Investigated representative LLM agent frameworks through literature review, open-source implementation analysis, and architectural comparison.
- Prepared technical evaluation reports summarizing empirical findings to support internal model selection and agent system optimization.

Technical Skills

- Programming Languages:** Python, Java, C/C++, SQL, JavaScript, Cypher
- AI:** LLMs, AI Agents, LLM Evaluation, Prompt Engineering
- Databases:** Neo4j, MySQL, PostgreSQL, Redis
- Software & Tools:** Spring Boot, RESTful APIs, Docker, Git, Linux